Correcting Selection Bias in Wearable Data: A Mass Imputation Framework for U.S. Physical Activity Surveillance

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Abstract

National health surveillance lacks objective, population-level benchmarks for specific movement metrics such as daily step counts and activity calories because surveys like the National Health and Nutrition Examination Survey (NHANES) do not collect wearable sensor data. While volunteer cohorts such as the All of Us (AoU) Research Program provide high-resolution Fitbit data, they are subject to significant selection bias. This study addresses this gap by implementing a mass imputation framework designed for correcting selection bias and transporting objective activity metrics from the AoU volunteer cohort ($n = 15,313$) to a representative NHANES population ($n = 4,914$) to estimate national mean physical activity levels. Two primary frameworks were compared to determine the most effective approach for correcting selection bias. The Standard Mass Imputation (MI) framework utilizes the entire available volunteer pool as a source pool, whereas the Matched Mass Imputation (MMI) framework employs 1:1 nearest-neighbor matching to refine the source pool to only the most demographically similar volunteers. Individual point estimates were first generated from five distinct predictive models, including Linear Regression, XGBoost, Random Forest, Neural Networks, and Generalized Additive Models. These results were then integrated into a consensus estimate that accounts for both the model-specific point estimates and their associated 95% confidence intervals derived from a dual-resampling bootstrap procedure. The performance of the framework was further validated through a sub-domain analysis across age, race, and gender categories. The results identified stable national benchmarks for daily steps and activity calories after correcting for the inherent healthy-volunteer bias. Under the Standard MI framework, the national average was estimated to be between 8,124 and 8,289 daily steps and 1,161 to 1,180 activity calories. The MMI framework significantly enhanced stability and narrowed the machine learning consensus for both metrics to a precise range of 8,236 to 8,263 steps and 1,170 to 1,173 calories. Notably, the analysis revealed a reverse-selection bias in older populations (65+) where corrected national estimates exceeded raw volunteer averages. This suggests that wearable-based studies may underrepresent active, community-dwelling seniors. Furthermore, a compensatory effect was observed where the representative population's calorie expenditure remained comparable to volunteers despite lower step counts due to a higher average body mass. This research provides a scalable, equitable blueprint for establishing objective national health benchmarks by successfully correcting selection bias through principled data integration.

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Chapter 1

Introduction

1.1 Background and motivation

Large-scale health datasets play a central role in monitoring population health and informing public policy in the United States. Among these, the National Health and Nutrition Examination Survey (NHANES) is widely regarded as the gold standard for nationally representative health surveillance. NHANES employs a complex, multistage probability sampling design that incorporates stratification, clustering, and oversampling of key demographic groups to ensure representativeness of the civilian, non-institutionalized U.S. population (CDC, 2024). By applying survey weights that account for unequal selection probabilities and nonresponse, researchers can generalize NHANES findings to the national level with well-characterized uncertainty.

In contrast, the All of Us (AoU) Research Program represents a newer case in population health research, emphasizing large-scale, longitudinal data collection from volunteer participants (AoU Research Hub, 2024). AoU integrates electronic health records, survey responses, physical measurements, biospecimens, and, through its "Bring Your Own Device" (BYOD) initiative, high-resolution wearable sensor data such as Fitbit-derived daily step counts and activity-related energy expenditure. While the size and clinical richness of the AoU cohort provide substantial analytical advantages, the absence of a probability-based sampling design introduces selection bias (AoU Research Program, 2024). Participants who volunteer for AoU and contribute wearable data tend to differ systematically from the general population in terms of health status, socioeconomic characteristics, and health-related behaviors, limiting the validity of simple population-level conclusions.

A critical limitation arises in the data structure of the August 2021–August 2023 NHANES cycle. Although NHANES provides extensive biometric, laboratory, and clinical measurements, it does not include device-based physical activity metrics such as average daily steps or activity calories burned. Instead, physical activity in this cycle is characterized through self-reported questionnaires that focus on categorical exercise intensity and sedentary duration. While these measures serve as important predictors within the modeling framework, they do not capture the specific metrics of interest for this study, namely the objective, continuous step counts, and calorie data provided by modern wearables. While NHANES has previously deployed accelerometers in earlier cycles, these data are not available in the current cycle and are not directly comparable to modern wrist-worn wearable devices. At the same time, the All of Us BYOD program captures precisely the sensor-based activity measures absent from NHANES, offering an opportunity to supplement national health surveillance with objective behavioral data. However, using AoU wearable data alone to estimate national physical activity levels would ignore the program's volunteer-based selection mechanism and could lead to biased estimates. As a result, researchers face a fundamental data integration challenge: how to leverage the measurement precision of wearable sensor data while retaining the population representativeness required for valid national inference.

This project is motivated by the need to address this challenge using principled statistical methods. By integrating the nationally representative structure of NHANES with the objective physical activity measurements available in the All of Us cohort, this study aims to generate population-level estimates of average daily steps and activity calories that are both demographically representative and biologically grounded. Establishing such estimates is essential for public health surveillance, as national benchmarks for physical activity are used to monitor trends, identify disparities across demographic groups, and inform initiatives such as Healthy People 2030. Furthermore, this study motivates the use of advanced machine learning architectures to capture the complex and non-linear relationships between demographic, anthropometric, and behavioral predictors and physical activity. By employing ensemble methods and non-linear frameworks, this project aims to create more accurate population estimates that properly account for the movement patterns of all demographic groups.

1.2 Review of Relevant Studies

Historically, population-level health studies relied almost exclusively on validated self-report instruments, such as Physical Activity Questionnaires (PAQs), due to the high cost and logistical complexity of deploying measurement devices in large cohorts. However, research by Prince et al. (2008) and others has consistently shown a low-to-moderate correlation between self-reported activity and device-based measurements, with individuals frequently overestimating their step counts. The shift toward device-based measurement via accelerometry in national surveys began with the NHANES 2003-2006 cycles, but cost constraints often prevent the continuous collection of these metrics in every cycle.

When essential data is difficult or expensive to collect, researchers often rely on statistical methods to fill these information gaps. Researchers have shifted away from traditional missing data treatments toward sophisticated data integration. Early literature often relied on Listwise Deletion, which is now recognized as potentially problematic for downstream statistical inference. As Pepinsky (2018) notes, listwise deletion can result in a loss of statistical power and introduces selection bias when data is not Missing Completely at Random (MCAR). Similarly, Lohr and Raghunathan (2017) emphasize that simply combining disparate data sources without accounting for different selection mechanisms and measurement errors can lead to biased population estimates. This necessitates a formal framework to bridge the NHANES 2021–2023 cycle with external sensor-based cohorts.

Modern research favors mass imputation to bridge the gap between representative probability surveys and large-scale external data sources like the All of Us Fitbit cohort. This framework, established by Yang et al. (2021), utilizes shared auxiliary variables to transport objective outcomes across datasets. The existing literature primarily distinguishes between two approaches: Standard Mass Imputation (MI) and Matched Mass Imputation (MMI). Kim et al. (2021) utilized a standard MI framework, which assumes that the relationship between predictors and activity is transportable under the "strong ignorability" assumption. However, Flood and Mostafa (2025) suggests that standard MI may remain susceptible to selection bias if the volunteer cohort differs significantly from the representative sample. They propose MMI as a refinement, matching observations based on a distance metric to ensure the training subset more closely mirrors the national representative sample.

Beyond point estimation, a significant concern in the literature is maintaining the authentic biological distribution and variability of behavioral metrics. Leiby and Ahner (2023) argue that relying on single, mean-based predictive architectures can lead to "flat" averages that fail to capture the full heterogeneity of the target population. They suggest that standard parametric approaches often underestimate the true spread of activity, which necessitates more robust variance estimation to recover the underlying population structure. Furthermore, Chen et al. (2025) emphasize that when integrating disparate datasets, traditional variance estimation can be insufficient, requiring more sophisticated frameworks to ensure that resulting confidence intervals remain statistically sound.

While recent work has established mass imputation as a principled framework for integrating probability and non-probability samples, relatively few studies have applied these techniques to high-variance behavioral sensor data such as physical activity. Moreover, direct comparisons between Standard Mass Imputation and matching-based refinements remain limited in applied public health contexts. This project builds on existing methodology to evaluate the practical performance of these approaches when integrating wearable device data into a national survey framework.

1.3 Problem Statement

Public health surveillance has yet to establish a national physical activity benchmark that fully integrates the objective precision of modern wearable devices into a representative framework. While NHANES provides a nationally representative framework for assessing American health, the August 2021–August 2023 cycle lacks high-resolution sensor data, such as average daily steps and average activity calories burned. Conversely, while the AoU Research Program captures this exact data through its Fitbit BYOD program, the cohort is a non-probability volunteer sample subject to selection bias, meaning its participants may not accurately reflect the physical activity levels of the general U.S. population. Consequently, researchers are faced with a data trade-off. They must choose between a sample that is representative but lacks the variables of interest (NHANES) or a sample that has the variables but is limited by selection bias (AoU).

If the biased All of Us data were used alone to make national claims, it would likely result in inaccurate estimates of physical activity. Wearable users possess profiles that differ systematically from the general population, and it is unlikely that this selection bias is uniform across all demographic groups. This project addresses this challenge by implementing and comparing two primary integration frameworks, Standard Mass Imputation (MI) and

Matched Mass Imputation (MMI), which are specifically designed to correct for these non-probability selection mechanisms. Rather than relying on a single model, this study identifies a consensus range across multiple machine learning architectures to adjust the volunteer data so that it reflects a nationally representative profile. By evaluating how different frameworks handle these systemic differences, the study establishes more reliable national benchmarks for daily steps and activity calories. Finally, the resulting estimates are examined to ensure they maintain biological plausibility after the bias-correction process has been applied.

The primary research question is: "How can mass imputation frameworks be utilized to correct for selection bias when transporting daily step and activity calorie data from the All of Us volunteer cohort to the nationally representative NHANES population, and how does model complexity impact the stability of these estimates?" To address this question, the project pursues the following objectives:

1. Selection Bias: To what extent do demographically adjusted national estimates for daily steps and calories differ from raw volunteer averages when evaluated by age, race, and gender?

2. Model Complexity: How do non-linear machine learning models such as XGBoost, Random Forest, Neural Networks, and GAMs compare to traditional linear regression in identifying stable consensus ranges for population-level movement and energy expenditure?

3. Framework Comparison: Does the Standard Mass Imputation framework or the Matched Mass Imputation framework provide more biologically plausible results when evaluating the step counts and calorie expenditure of demographic groups?

The remainder of this report is organized as follows. Chapter 2 describes the data sources, preprocessing procedures, and harmonization strategy used to align NHANES and All of Us data. This is followed by a detailed description of the Standard Mass Imputation and Matched Mass Imputation frameworks, alongside the linear and non-linear machine learning architectures used to account for model complexity. Chapter 3 presents the results of the study, including the bias-corrected population-level estimates, demographic sub-domain analyses, and a comparative evaluation of model stability and consensus across both integration frameworks. Chapter 4 provides a discussion of the findings, evaluates the biological plausibility of the estimates, and explores the implications of these results for national public health surveillance. The report concludes with an assessment of the limitations of the study and suggestions for future research in data transportability and the correction of selection bias in volunteer-based sensor cohorts.

Chapter 2

Methodology

2.1 Data Sources and Harmonization

This study integrates data from two primary sources: the National Health and Nutrition Examination Survey (NHANES) and the All of Us (AoU) Research Program. The NHANES August 2021–August 2023 cycle serves as the nationally representative cohort ($n = 4,914$ adults), providing a probability-based sample of the non-institutionalized U.S. population. In contrast, the All of Us "Registered Tier Dataset v8" provides a large-scale volunteer donor pool ($n = 15,313$), contributing high-resolution Fitbit-derived records for daily steps and activity calories that are absent from the NHANES cohort.

Preprocessing and Quality Control

The preprocessing pipeline transformed raw, diverse records into a unified format suitable for mass imputation. For the AoU cohort, sensor records were restricted to valid days, defined as at least 600 minutes of heart rate-monitored wear time. Additionally, participants were required to have a minimum of seven valid days to be included. To ensure physiological plausibility, extreme outliers were removed. Average daily steps were restricted to a range of 1,000 to 50,000, while activity calories were filtered to remain between 50 and 3,000. For the NHANES cohort, the sample was restricted to adults aged 18 years and older with valid, non-zero Mobile Examination Center weights to preserve the integrity of the national probability sample.

Variable Harmonization

Harmonization was conducted to align the feature space (X) across both cohorts, ensuring that models trained on the volunteer sample could be validly transported to the national sample. While 12 initial candidates were considered, the feature set was refined based on a missingness audit to maximize the complete-case sample size required for stable integration. Consequently, several variables with high rates of missing values were excluded, including race, education level, smoking status, alcohol consumption, and heart rate. Additionally, Body Mass Index was removed to avoid multicollinearity with height and weight. These exclusions prioritized a high-quality data profile over maximal covariate inclusion, ensuring that both the training and representative cohort shared a stable and consistent set of predictors.

Table 2.1: Final Harmonized Feature Space (X)

Variable	**Type**	**Harmonization Strategy**
Age	Continuous	Verified in years and restricted to adults 18 and older.
Gender	Binary	Recoded into a unified factor for Male and Female.
Height	Continuous	Verified and standardized to centimeters.
Weight	Continuous	Standardized to kilograms and filtered for realistic ranges.
Sedentary Minutes	Continuous	Filtered to capture realistic awake-time sedentary behavior.
Active Minutes	Continuous	Verified for a physiologically plausible daily range.

2.2 Mass Imputation Frameworks

The core of this research involves transporting physical activity outcomes (Y), specifically daily steps and activity calories, from the volunteer All of Us (AoU) cohort to the representative NHANES sample. This process is conducted using two primary integration methods: Standard Mass Imputation (MI) and Matched Mass Imputation (MMI). To ensure these methods produce valid national estimates, the integration frameworks must satisfy specific statistical conditions regarding data transportability and selection mechanisms.

Identification and Transportability Assumptions

Let S_A denote the probability sample (NHANES) and S_B denote the non-probability volunteer cohort (AoU). To ensure the validity of transporting predictive relationships from S_B to S_A, the framework relies on the assumption of Strong Ignorability. This posits that the sampling mechanism for the volunteer cohort is independent of the activity outcomes, given a shared set of auxiliary variables X:

$$Y \perp\!\!\!\perp S \mid X.$$

In addition to ignorability, the framework assumes Positivity, which requires that every demographic profile present in the representative NHANES cohort has a non-zero probability of being represented within the volunteer source pool. Under these combined assumptions, the conditional expectation $E[Y|X]$ is transportable, allowing a model $\hat{f}_{AoU}$ trained on the volunteer sample to serve as a valid proxy for the unobserved outcomes in the national sample.

The General Mass Imputation Estimators

The transport process occurs in two stages: individual prediction followed by population-level aggregation.

1. Individual Prediction. For each participant i in the representative NHANES sample (S_A), an individual-level predicted outcome $\hat{y}_i$ is generated using the model architecture $\hat{f}_{AoU}$ trained on the volunteer cohort:

$$\hat{y}_i = \hat{f}_{AoU}(x_i).$$

2. Population Point Estimate. To generate the final national benchmark, these individual predictions are scaled to the national level using NHANES complex survey weights (w_i). These weights are based on the inverse inclusion probability, π_i^{-1}, which accounts for the unequal selection probabilities inherent in the NHANES multistage sampling design:

$$\pi_i^{-1} = \frac{1}{\Pr(i \in S)}.$$

The national point estimate $\hat{\theta}$ is then defined by the following weighted estimator:

$$\hat{\theta} = \frac{\sum_{i \in S_A} w_i \cdot \hat{y}_i}{\sum_{i \in S_A} w_i},$$

where $\hat{\theta}$ represents the final national point estimate for the chosen physical activity metric, such as mean daily steps or activity calories , S_A denotes the set of NHANES participants included in the analytical sample , and w_i represents the recalibrated survey weight for participant i. The term $\hat{y}_i$ is the individual-level predicted outcome for participant i generated by the predictive model , while $\Pr(i \in S)$ represents the probability that individual i was selected for inclusion in the probability sample S. In this formula, the denominator $\sum w_i$ represents the total estimated U.S. non-institutionalized population , ensuring the final mean accurately reflects national physical activity levels.

Standard Mass Imputation (MI)

The Standard MI framework serves as the baseline integration technique. It utilizes the full All of Us volunteer cohort ($n = 15,313$) to train the predictive function $\hat{f}_{AoU}$. This approach assumes that the large size and diversity of the volunteer pool are sufficient to capture the relationship between anthropometric and demographic features and physical activity, which is then transported to the representative NHANES sample using the estimators defined in the previous section.

Matched Mass Imputation (MMI)

The MMI framework introduces a preprocessing matching step to ensure Positivity and distributional overlap between the two datasets. By effectively aligning the source covariate profile with that of the representative NHANES sample, this step ensures that the training data contains participants who are demographically comparable to the representative cohort. For source selection, a 1:1 nearest-neighbor matching procedure is employed to identify a representative training subset ($n = 4,914$). A unique counterpart $j(i)$ from the volunteer cohort (S_B) is identified for every NHANES participant (i) by minimizing the Euclidean distance, or L_2 norm, across the shared feature space:

$$j(i) = \arg\min_{j \in S_B} \sqrt{\sum_{k=1}^{p} (x_{A,i,k} - x_{B,j,k})^2}.$$

In this formula, the arg min operator identifies the specific index j within the volunteer cohort S_B that results in the minimum calculated distance for the i-th participant in S_A. Consequently, $j(i)$ represents the specific participant in the volunteer cohort S_B that serves as the nearest-neighbor match for individual i in the representative NHANES sample (S_A). The variables $x_{A,i,k}$ and $x_{B,j,k}$ represent the values of the k-th shared feature for those individuals, summed across all p harmonized predictors. Following the selection of this representative source pool, targeted transport is performed by training the predictive function $\hat{f}_{AoU}$ exclusively on this matched subset. By restricting the training data to individuals who mirror the national distribution, the MMI framework aims to produce model coefficients that are less influenced by the selection bias of the broader volunteer sample prior to generating the final weighted national estimates.

2.3 Predictive Modeling Architectures

To address the potential non-linearity of physical activity data, this study implements a diverse suite of predictive architectures. By utilizing multiple modeling approaches ranging from traditional parametric baselines to ensemble-based machine learning, the research aims to identify a stable consensus for national activity estimates while specifically correcting for selection bias within the volunteer-based dataset. Each of these architectures was evaluated for its ability to generate stable predictions across both data integration frameworks, ensuring that the correction for selection bias remained consistent regardless of model complexity. To ensure reliable performance, all models were configured with parameters appropriate for high-dimensional behavioral data to facilitate stable convergence across both the Standard and Matched Mass Imputation workflows.

Baseline Linear Regression
Multiple linear regression serves as the baseline model. While its simplicity offers high interpretability, it assumes a constant relationship between the covariates and physical activity. The baseline model is defined as:

$$y_i = \beta_0 + \beta_1 x_{i,1} + \cdots + \beta_k x_{i,k} + \epsilon_i.$$

In this formulation, $x_{i,k}$ represents the shared covariates, which include features such as age, gender, height, weight, active minutes, and sedentary minutes. The term β_k represents the fixed coefficient estimated for each of these features from the source cohort. To estimate these coefficients,the model minimizes the Ordinary Least Squares (OLS) objective function with respect to the parameter vector β:

$$\min_{\beta} \sum_{i=1}^{n} (y_i - x_i^T \beta)^2.$$

In this objective function, n denotes the total number of participants in the source pool, y_i is the observed physical activity outcome for individual i, and x_i^T represents the transposed vector of explanatory variables for that individual. This model is included to evaluate whether traditional statistical methods are sufficient for transportability or if the complexity of sensor data necessitates more flexible modeling.

Generalized Additive Models (GAM)

The GAM was selected to provide a bridge between linear and machine learning approaches. By using smoothing functions, the GAM allows for non-linear relationships, such as the observed non-linear decline of physical activity with advancing age, while maintaining statistical transparency. The structural equation for this model is defined as:

$$y_i = \beta_0 + \text{gender}_i + \sum_{j=1}^{5} f_j(x_{i,j}) + \epsilon_i.$$

Within this modeling structure, each f_j represents a unique thin-plate regression spline, which is a flexible basis function used to estimate the smooth effect of a continuous predictor. The term $x_{i,j}$ denotes the value of the j-th harmonized covariate specifically age, height, weight, active minutes, and sedentary standards for individual i. To prevent overfitting and maintain numerical stability, the basis dimension for all smoothing functions was constrained to $k = 5$. This allows each physical and behavioral feature to have its own non-linear relationship with the outcome while ensuring that the functional complexity of the curves remains biologically plausible. This model is essential for capturing physiological trends that a strictly linear framework might oversimplify or miss entirely.

Tree-Based Ensemble Models

To capture complex and non-linear interactions between features that may be missed by additive models, this study utilizes Random Forest (RF) and Extreme Gradient Boosting (XGBoost). These models share a common additive structure where the final prediction $\hat{y}_i$ for a given participant is the sum of K base learners:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i), \quad f_k \in \mathcal{F}.$$

In this equation, f_k represents an individual decision tree, and $\mathcal{F}$ denotes the functional space of all possible regression trees. Each tree f_k maps the input features x_i to a specific score based on the learned decision rules. While both models utilize ensembles of trees, they differ significantly in their optimization and training strategies.

Random Forest utilizes a bagging (bootstrap aggregating) approach to reduce variance. It builds K independent, deep decision trees in parallel, each trained on a different bootstrap sample of the volunteer cohort. The final prediction is the simple average of these K trees, which helps to mitigate the influence of outliers and reduce overfitting. In contrast, the XGBoost framework utilizes a gradient boosting mechanism. Instead of building trees independently, it builds them sequentially, where each new tree is designed to minimize the residual errors of the previous ensemble. This is achieved by minimizing a regularized objective function:

$$\text{obj} = \sum_i L(y_i, \hat{y}_i) + \sum_k \Omega(f_k).$$

In this formulation, L represents the training loss function (Mean Squared Error) which measures the distance between the observed physical activity outcome y_i and the prediction $\hat{y}_i$. The term Ω is the regularization component used to penalize model complexity and ensure the transportability of the coefficients:

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda||w||^2,$$

where T denotes the number of leaves in the tree and w represents the vector of leaf weights. The parameter γ acts as a complexity cost for adding a new leaf, while λ serves as the L_2 regularization coefficient. By minimizing this objective function, XGBoost incorporates the specific penalties necessary to maintain numerical stability when dealing with high-dimensional behavioral data.

Neural Network (NN)

A multi-layer perceptron architecture is employed to model high-dimensional relationships and cross-variable interactions through non-linear transformations. The structural flow of the network, from the input features x to the hidden layer and finally the output prediction $\hat{y}$, is defined by the following equations:

$$h(x) = \sigma(W_1 x + b_1),$$

$$\hat{y} = W_2 h(x) + b_2.$$

In these formulas, W_1 and W_2 represent the weight matrices for the hidden and output layers, respectively, which determine the strength of the connections between neurons. The terms b_1 and b_2 denote the bias vectors, which allow the activation function to be shifted to better fit the data. The operator σ represents the activation function, such as ReLU (Rectified Linear Unit) or Sigmoid, which introduces the necessary non-linearity to capture complex behavioral patterns.

For this study, a linear activation function was applied at the output layer to generate continuous numerical estimates for daily steps and activity calories. To prevent overfitting and ensure the weights do not grow disproportionately large, a weight decay penalty was incorporated into the training process. This ensures numerical stability and maintains the generalizability of the model when transporting predictions from the All of Us volunteer cohort to the national NHANES sample.

2.4 Statistical Inference and Variance Estimation

The final phase of the methodology involves quantifying the uncertainty inherent in the transportability process. Because this study integrates a non-probability volunteer cohort with a complex probability survey, the variance estimation must account for both sampling error and predictive modeling error. This requires an inferential framework that treats the data integration as a process driven by probabilistic variability, ensuring that the resulting 95% Confidence Intervals (CI) capture the cumulative uncertainty required for valid population-level estimates.

Dual-Resampling Bootstrap Variance

To provide robust 95% Confidence Intervals (CI), this study implements a Dual-Resampling Bootstrap procedure. This approach captures the cumulative variability by accounting for sampling variance within the representative NHANES sample (S_A) as well as predictive uncertainty originating from the volunteer training cohort (S_B). The procedure follows a formal four-step mathematical algorithm executed over $B = 100$ replicates:

1. **Resampling:** For each replicate $b \in \{1, \ldots, B\}$, draw a bootstrap sample S_A^* from NHANES and a bootstrap sample S_B^* from the volunteer cohort, both using non-parametric resampling with replacement.

2. **Train Bootstrap Model:** Train the predictive model architecture $\hat{f}^*$ exclusively on the resampled volunteer data S_B^*.

3. **Project and Aggregate:** Generate predictions for the resampled national sample and calculate the bootstrap replicate estimate:

$$\hat{\theta}^* = \frac{\sum_{i \in S_A^*} w_i^* \cdot \hat{f}^*(x_i^*)}{\sum_{i \in S_A^*} w_i^*}.$$

4. **Construct Distribution and Final Estimates:** Repeat the process to generate a distribution of $B = 100$ replicate estimates $\{\hat{\theta}^{*(1)}, \hat{\theta}^{*(2)}, \ldots, \hat{\theta}^{*(B)}\}$. The final reported point estimate, $\hat{\theta}_{final}$, is defined as the mean of this bootstrap distribution:

$$\hat{\theta}_{final} = \frac{1}{B} \sum_{b=1}^{B} \hat{\theta}^{*(b)}.$$

Confidence Interval Construction

The final 95% CIs were constructed using the Percentile Method, which determines the range of the bootstrap distribution without requiring an assumption of normality. For a significance level of $\alpha = 0.05$, the interval is defined by the $(\alpha/2)$ and $(1 - \alpha/2)$ quantiles of the bootstrap distribution:

$$95\% \text{ CI} = [\hat{\theta}^*_{(0.025)}, \hat{\theta}^*_{(0.975)}].$$

In this notation, $\hat{\theta}^*_{(0.025)}$ and $\hat{\theta}^*_{(0.975)}$ represent the 2.5th and 97.5th percentiles of the ordered bootstrap estimates, respectively. This method provides a non-parametric measure of total uncertainty, encompassing both the sampling design and the algorithmic variance of the mass imputation framework.

2.5 Sub-Domain Analysis

A primary objective of this study is to evaluate the performance of the MI and MMI frameworks across diverse demographic sub-populations. This analysis is critical because selection bias in volunteer-based cohorts often varies significantly by age, race, and gender, which can lead to disparate predictive accuracy across demographic groups. By moving from a national aggregate to a granular sub-domain level, this study assesses whether the data integration process effectively mitigates these localized biases or if the model's performance remains sensitive to the socio-demographic composition of the training data.

For each demographic sub-domain d, a group-specific mean estimate is calculated to represent the national average for that population. Because the exclusion of individuals with missing demographic identifiers can inadvertently alter the survey's original population totals, a post-stratification weight adjustment is employed. This process recalibrates the NHANES survey weights w_i^* to ensure that the individuals within each sub-domain accurately represent their respective proportion of the U.S. population. The weight adjustment for the sub-domain is defined as follows:

$$w_i^* = w_i \times \left(\frac{\sum_{j \in S_{Total}} w_j}{\sum_{k \in S_{Sub}} w_k} \right).$$

In this expression, S_{Total} represents the full representative population for the sub-domain and S_{Sub} represents the subset of respondents with non-missing demographic identifiers. Once these weights are established, the sub-domain mean $\hat{\theta}_d$ is defined by the following estimator:

$$\hat{\theta}_d = \frac{\sum_{i \in S_{A,d}} w_i^* \cdot \hat{y}_i}{\sum_{i \in S_{A,d}} w_i^*}.$$

In this formula, $\hat{\theta}_d$ is the estimated national average for sub-domain d. The term $S_{A,d}$ denotes the specific NHANES demographic subset, $\hat{y}_i$ represents the predicted activity outcome for individual i, and w_i^* refers to the post-stratified survey weight adjusted to maintain population representativeness despite missing identifiers.

Uncertainty for these sub-domain estimates was quantified using the Dual-Resampling Bootstrap framework described in the previous section[cite: 13]. This ensures that the 95% Confidence Intervals account for both sampling and model-related variance at the demographic level. This methodological approach allows for the identification of reverse bias and other disparities, such as those observed in older adults and specific racial groups, that would otherwise be masked by population-wide averages.

Chapter 3

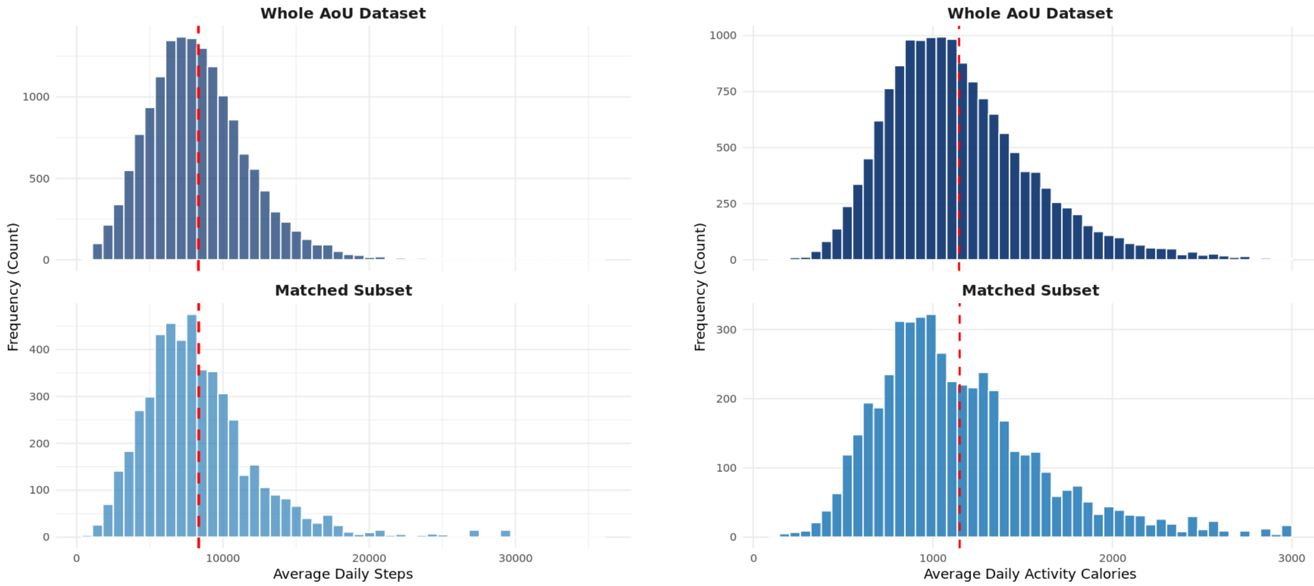
Results

3.1 Data Alignment and Distribution

Before imputation, the All of Us (AoU) volunteer pool ($n = 15,313$) was compared to the refined matched subset ($n = 4,914$) generated through the Matched Mass Imputation (MMI) framework. Histograms of daily steps and activity calories (Figure 3.1a) demonstrate that the matching process successfully preserved the natural variation and distribution shape of the original data.

As indicated by the dashed red lines in Figure 3.1, the central tendencies of both distributions remain closely aligned. This confirms that the MMI framework did not fundamentally distort the behavioral profile of the source pool. While the visual shift in the mean is subtle, the numerical analysis in Table 3.1 reveals a consistent upward correction. Specifically, mean daily steps in increased from 8,320 to 8,342, while mean activity calories increased from 1,146.2 to 1,149.5.

Notably, the standard deviation increased for both metrics, rising from 3,425.4 to 3,929.0 for daily steps and from 403.9 to 469.3 for activity calories. This 14.7% increase in variance suggests that the correction provided by the MMI framework is primarily found in the inclusion of a more diverse range of movement patterns. This consistent shift in both mean and variance indicates that the source pool most demographically aligned with the representative NHANES sample is slightly more active and exhibits greater behavioral diversity than the broader volunteer sample.



(a) Daily Steps (b) Activity Calories

Figure 3.1: Distributional comparison of physical activity outcomes between the full All of Us cohort and the representative matched source subset.

Table 3.1: Comparison of Mean Activity Metrics and Variability Between the Original AoU Pool and MMI Matched Subset

Metric	**Original AoU Dataset**	**Matched Subset**
Daily Step Count	8,320.0 (± 3,425.4)	8,342.0 (± 3,929.0)
Activity Calories	1,146.2 (± 403.9)	1,149.5 (± 469.3)

Note: Values are presented as Mean (± Standard Deviation).

3.2 Population-Level Estimates

The application of the Standard Mass Imputation (MI) framework to the full volunteer cohort revealed distinct shifts when transporting data to the representative NHANES sample. These estimates were compared against "Naive" volunteer averages, which were established at 8,317 daily steps and 1,146 activity calories. According to these primary results (Table 3.2), national estimates for mean daily steps were lower than volunteer averages across the flexible modeling approaches, such as XGBoost (8,124) and GAM (7,876). Conversely, activity calories showed a slight upward trend, with models converging between 1,161 and 1,167 calories. These findings highlight a divergence in metric shifts, where steps decreased while activity calories remained stable or slightly increased.

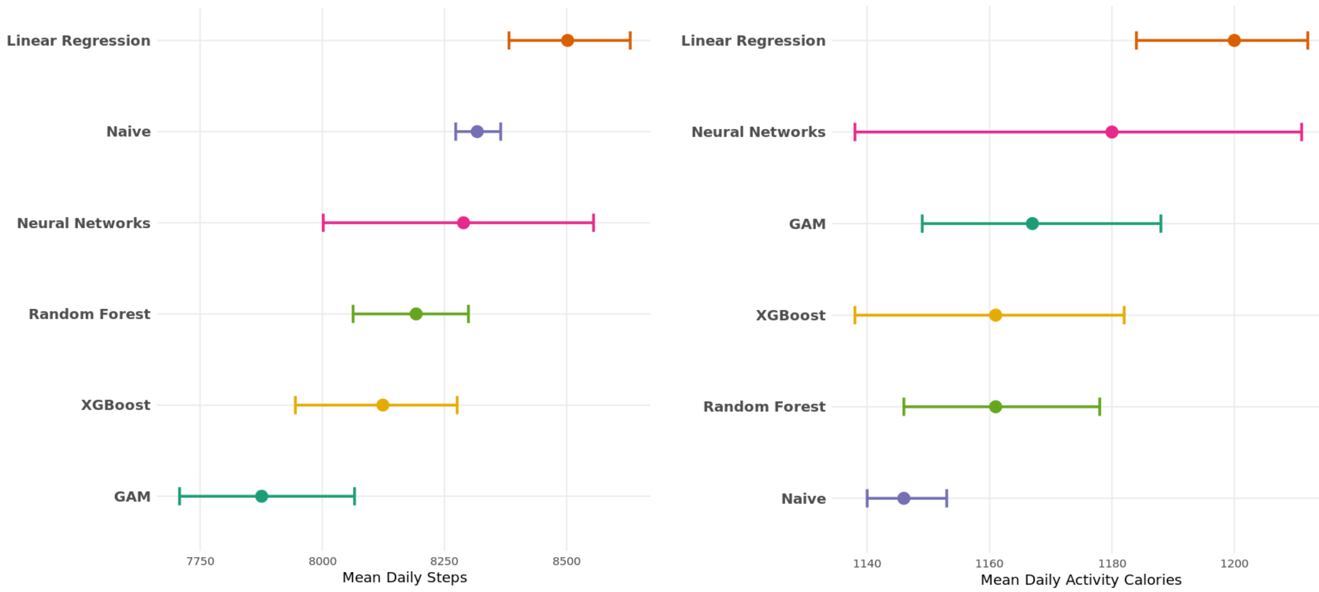


Figure 3.2: Transported population-level estimates for daily steps (left) and activity calories (right) across the Standard MI predictive architectures.

Table 3.2: Comparison of Transported National Estimates for Daily Steps and Activity Calories across MI Model Architectures

	Daily Steps		Activity Calories	
Model Architecture	**Mean**	**95% CI**	**Mean**	**95% CI**
Naive (Volunteer)	8,317	[8,273, 8,365]	1,146	[1,140, 1,153]
Random Forest	8,192	[8,063, 8,299]	1,161	[1,146, 1,178]
XGBoost	8,124	[7,945, 8,276]	1,161	[1,138, 1,182]
GAM	7,876	[7,708, 8,066]	1,167	[1,149, 1,188]
Neural Networks	8,289	[8,002, 8,555]	1,180	[1,138, 1,211]
Linear Regression*	8,502	[8,382, 8,630]	1,200	[1,184, 1,212]

*Identified as a model outlier.

Rather than relying on a single predictive value, these results establish a consensus range derived from the convergence of multiple non-linear architectures. For daily steps, while the naive volunteer average was 8,317, the complex models (XGBoost, Random Forest, and Neural Networks) reached a stable consensus between 8,124 and 8,289 steps. Linear Regression was identified as a model outlier in this framework, as it was the only architecture to produce an estimate (8,502) significantly exceeding the naive average. A similar consensus emerged for activity calories, where the non-linear models converged between 1,161 and 1,180 calories. The tight clustering of these distinct machine learning frameworks, coupled with the consistent deviation of the linear model, provides strong evidence for the robustness of the national estimates and the necessity of non-linear modeling in this context.

3.3 Sub-Domain Analysis and Health Disparities

As selection bias is not uniform across demographics, sub-domain analysis was conducted to identify how transportability affects specific groups. Across all sub-domains, the flexible modeling architectures (XGBoost, Random Forest, Neural Networks, and GAMs) demonstrated high concordance, converging on similar estimates. In contrast, the baseline linear regression model consistently acted as a directional outlier, frequently overestimating activity levels compared to these non-linear approaches.

Age Group Analysis

Analysis of the age sub-domains revealed a striking reversal of selection bias in older populations (Figure 3.3). While the 18–34 and 35–49 age groups saw significant downward adjustments in activity, the 65+ demographic exhibited a 'reverse bias' where national activity levels were higher than the Naive volunteer average. For daily steps within the 65+ group, the Naive average of 8,092 was adjusted upward to a transported estimate of 8,257 by the Neural Network and 8,173 by the Random Forest. This trend was even more pronounced in activity calories. In this case, the 65+ Naive average of 1,024 calories was corrected to a national estimate of 1,055 calories by both the GAM and Random Forest models. These results suggest that older adults who participate in digital health research may actually be less active than their peers in the general U.S. population.

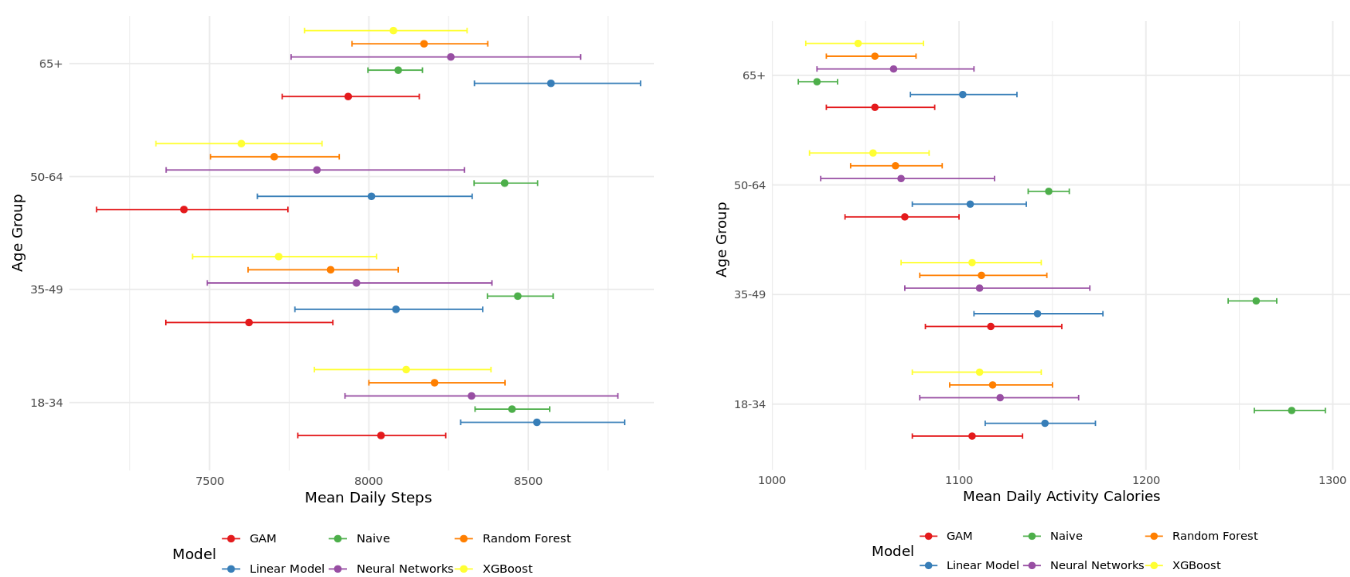


Figure 3.3: Transported population-level estimates for daily steps (left) and activity calories (right) stratified by age sub-domains.

Racial Sub-Domains

The framework revealed that selection bias varies significantly by race, correcting for the demographic imbalances inherent in the All of Us source pool (Figure 3.4). In the Black demographic, the models identified a consistent underestimation of activity. Specifically, the

Naive volunteer average of 8,041 steps was adjusted upward to a transported estimate of 8,323 by the Neural Network, while activity calories rose from a Naive average of 1,219 to 1,252. Conversely, the Hispanic demographic saw a substantial downward correction from a Naive average of 8,851 steps to a transported estimate of 8,247 via the GAM. For White participants, the models exhibited the highest stability, with most estimates remaining within 1% of the Naive baseline. This suggests that the "average volunteer" profile in volunteer-based digital health cohorts most closely mirrors the activity patterns of the White American demographic, while failing to capture the unique variations present in other racial groups.

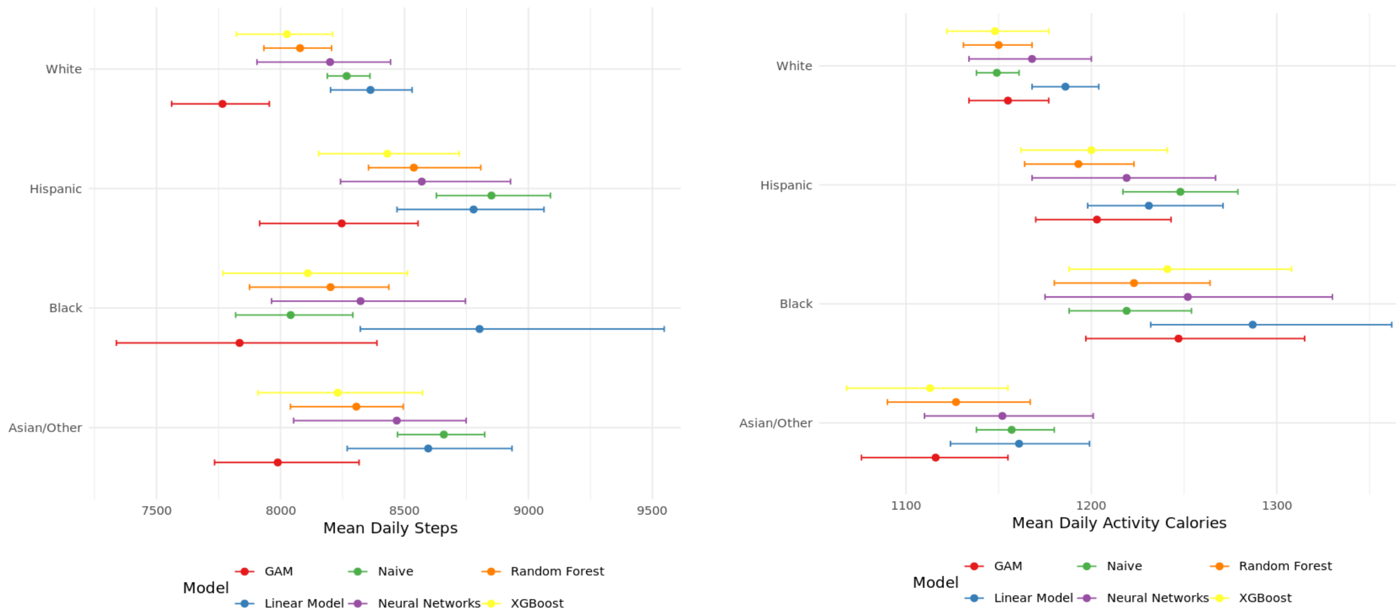


Figure 3.4: Transported population-level estimates for daily steps (left) and activity calories (right) stratified by racial sub-domains.

Gender-Based Shifts in Movement

Gender analysis highlighted distinct movement profiles that were previously masked by population-wide averages (Figure 3.5). Male participants saw the most dramatic downward adjustments in movement volume. Specifically, the Naive volunteer average of 8,845 steps was corrected to a transported estimate of 8,332 by XGBoost and 8,353 by Random Forest, suggesting a strong "Healthy Volunteer" bias among men in the study. Female participants also saw downward adjustments, though they were more concentrated in activity calories. For this group, the Naive average of 1,026 calories was refined by the models to a transported estimate between 982 (GAM) and 1,016 (Neural Networks).

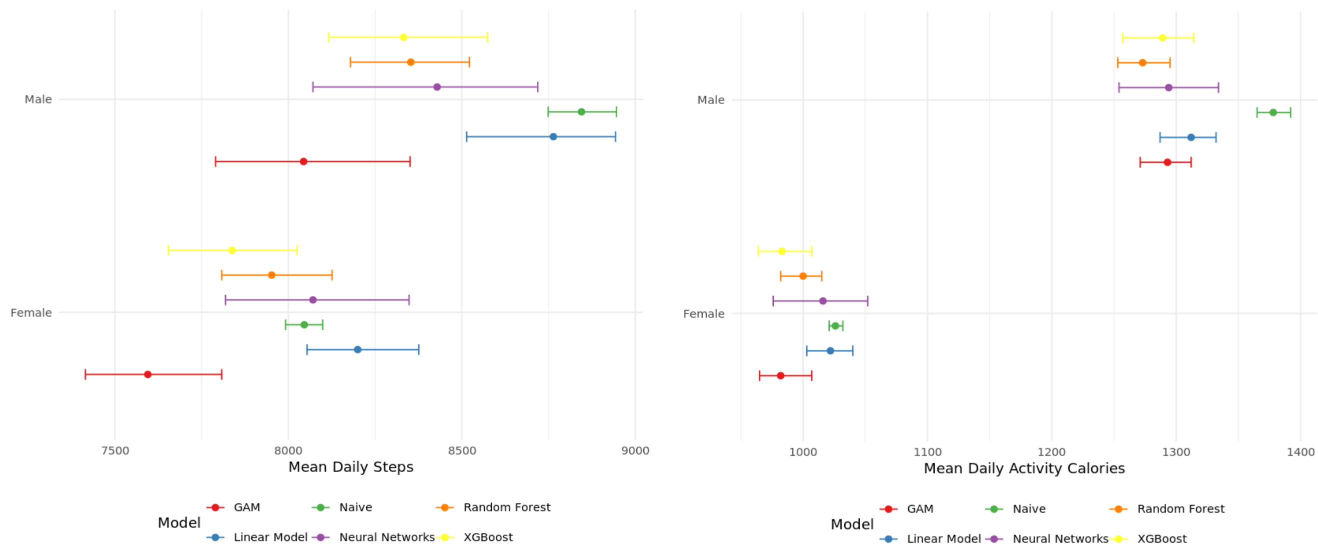


Figure 3.5: Transported population-level estimates for daily steps (left) and activity calories (right) stratified by gender sub-domains.

Ultimately, the sub-domain analysis across age, race, and gender confirms that the model consensus remains stable at these specific demographic levels. Across all categories, XGBoost, Random Forest, and Neural Networks demonstrated high concordance in the direction and magnitude of demographic shifts. This collective agreement allowed for the identification of the observed reverse bias in older adults and other disparities as consistent physiological trends rather than model-specific artifacts.

3.4 Framework Comparison: Standard MI vs. Matched MI (MMI)

To evaluate the robustness of the national estimates, this study compared the Standard Mass Imputation (MI) framework, which utilizes the full volunteer cohort ($n = 15,313$), against the Matched Mass Imputation (MMI) framework, which uses a demographically balanced subset ($n = 4,914$). This comparison identifies how restricting the source pool to the most demographically representative individuals influences the final transportability results.

Stability and Directionality of Population Estimates

At the population level, both frameworks produced a high degree of concordance in the direction of the shifts (Figure 3.6). For daily steps, the Standard MI models established a consensus range between 8,124 and 8,289 steps when excluding the linear and GAM outliers. The MMI framework produced a more conservative and significantly tighter consensus, with XGBoost, Random Forest, and Neural Networks converging almost exactly between 8,236 and 8,263 steps.

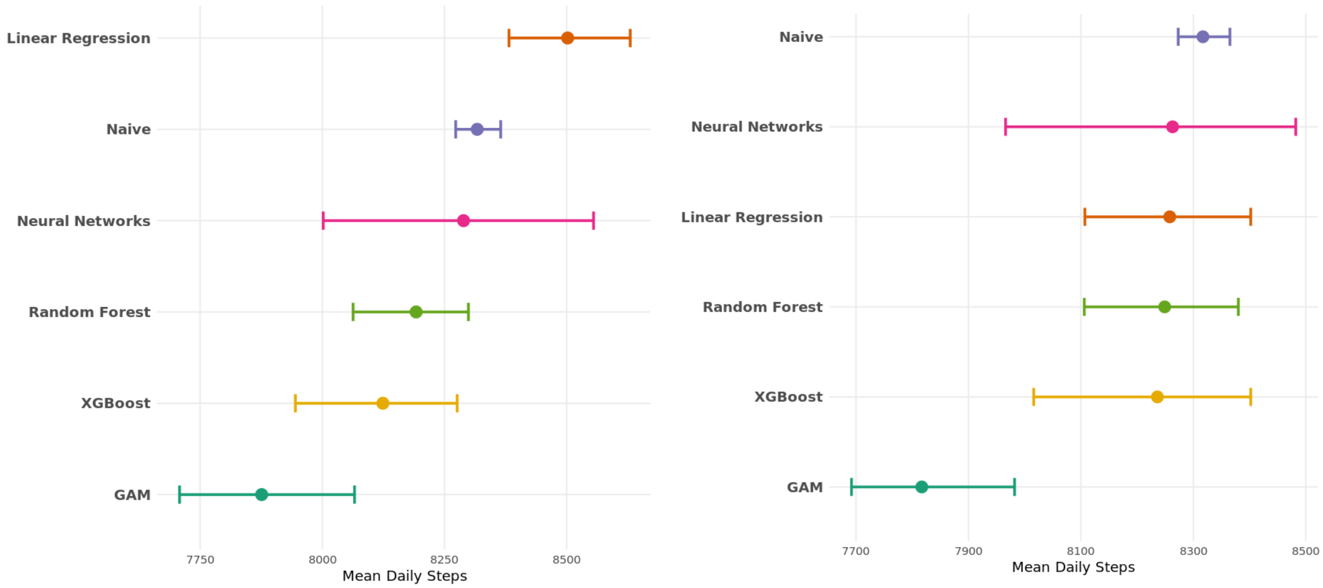


Figure 3.6: Transported daily step estimates comparing the Standard Mass Imputation (left) framework and the Matched Mass Imputation (right) framework across all modeling architectures.

A similar pattern emerged in the activity calorie domain (Figure 3.7). The Standard MI machine learning models converged on a range of 1,161 to 1,180 calories. Under the MMI framework, this consensus shifted slightly downward and tightened to a range of 1,154 to 1,169 calories. The fact that the machine learning models across both frameworks land within approximately 15 calories of each other despite using different source pools indicates a remarkably high level of stability in the national estimate.

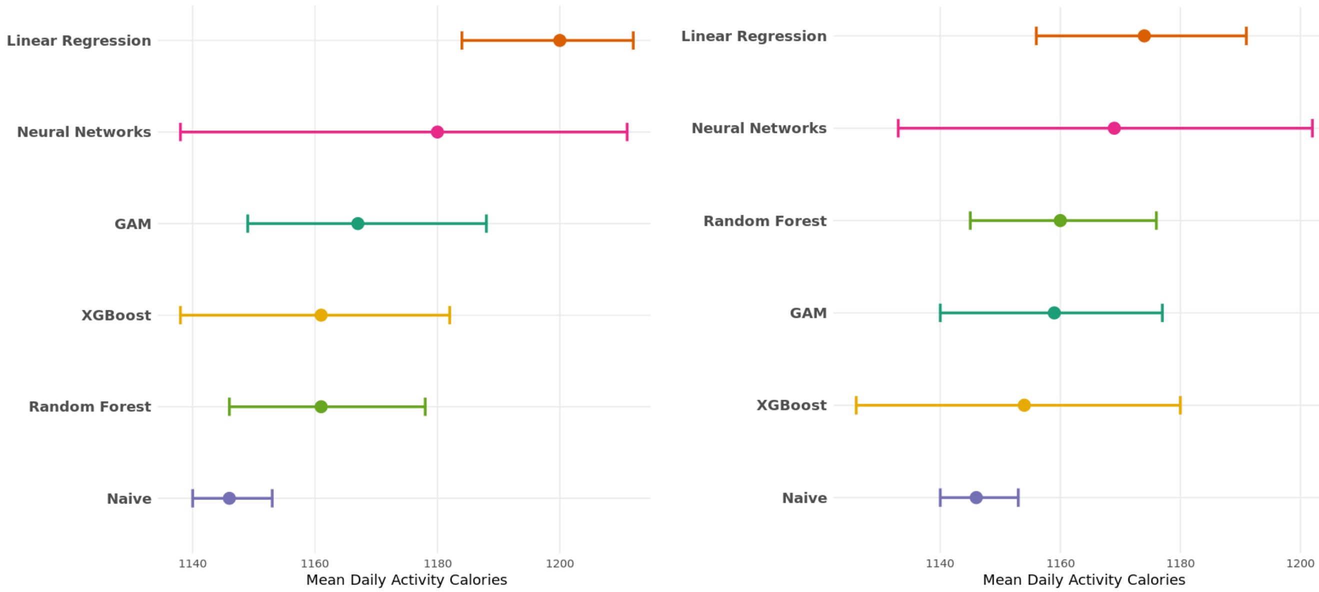


Figure 3.7: Transported activity calorie estimates comparing the Standard Mass Imputation (left) framework and the Matched Mass Imputation (right) framework across all modeling architectures.

Comparative Magnitude of Estimates

A consistent trend across nearly all models and sub-domains was that the MMI framework yielded higher estimates for daily steps compared to the Standard MI. At the population level (Figure 3.6), the MMI consensus for steps was higher than the Standard MI consensus by an average of approximately 100 steps. This pattern was also reflected in the age sub-domains (Figure 3.8), where the 65+ MMI transported estimate reached 8,249 steps compared to the Standard MI estimate of 8,173 steps. These results suggest that the refined matching process in the MMI framework effectively mitigates the over-correction of activity levels seen in the larger, less representative source pool.

In contrast, the calorie domain demonstrated a different behavior between the two frameworks (Figure 3.7). While the step counts increased under MMI, the calorie estimates remained remarkably stable or shifted slightly lower. The Standard MI machine learning models converged on a range of 1,161 to 1,180 calories, while the MMI consensus tightened to a range of 1,154 to 1,169 calories. This divergence suggests that restricting the source pool through demographic matching has a more pronounced upward effect on movement volume than on total energy expenditure.

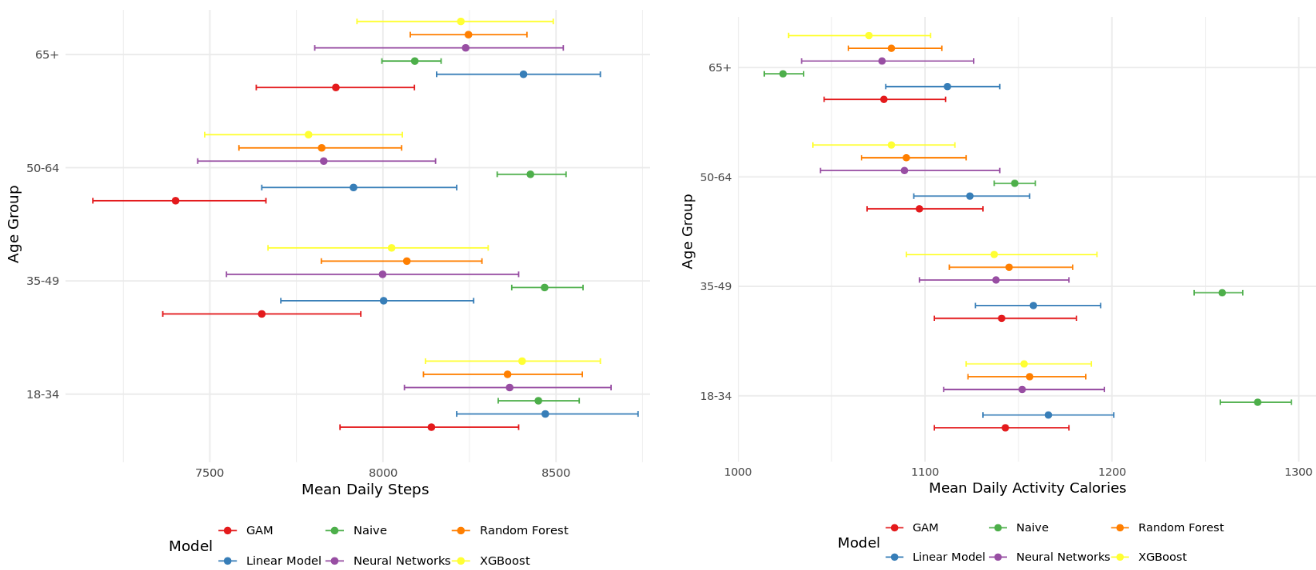


Figure 3.8: Transported age sub-domain estimates for daily steps (left) and activity calories (right) generated via the Matched Mass Imputation (MMI) framework.

Sub-Domain Reliability and Model Consensus

Both frameworks independently identified the same demographic-specific shifts. In the racial sub-domains (Figure 3.9), both MI and MMI shifted the Naive average for the Black demographic upward, reaching 1,252 and 1,241 calories, respectively. Similarly, in the gender analysis (Figure 3.10), both frameworks identified a healthy volunteer bias among males, with both models moving the Naive average of 8,845 steps significantly downward. The consistency in these directional shifts across frameworks, despite the difference in source pool size, indicates that the underlying drivers of selection bias are captured effectively by both methodologies.

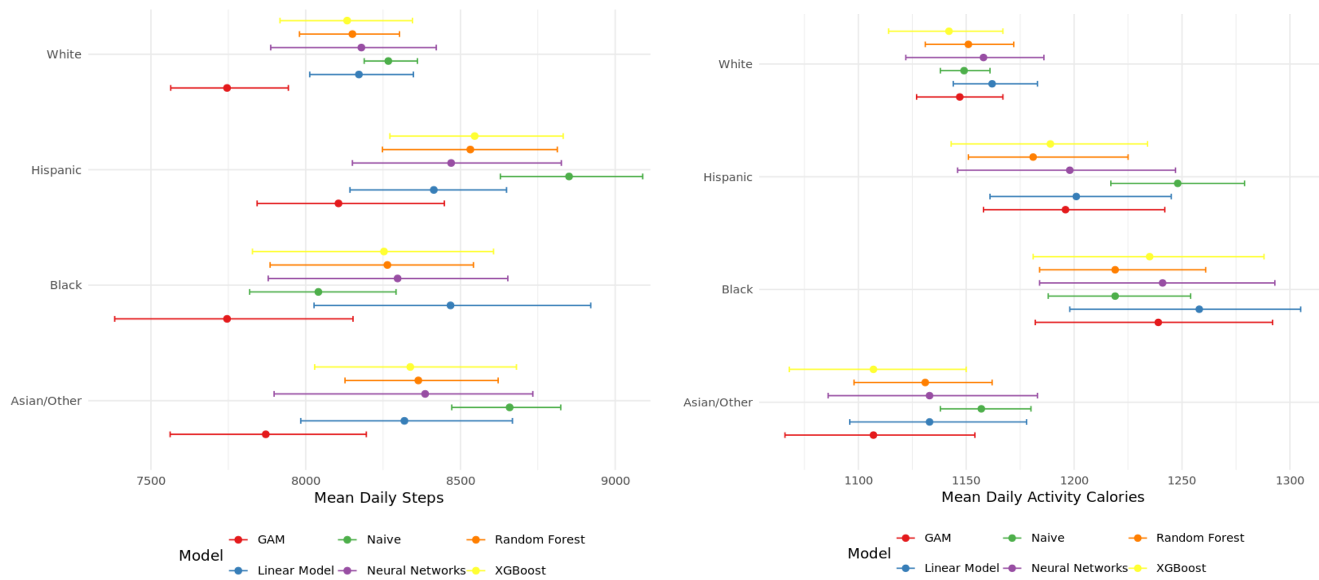


Figure 3.9: Transported racial sub-domain estimates for daily steps (left) and activity calories (right) generated via the Matched Mass Imputation (MMI) framework.

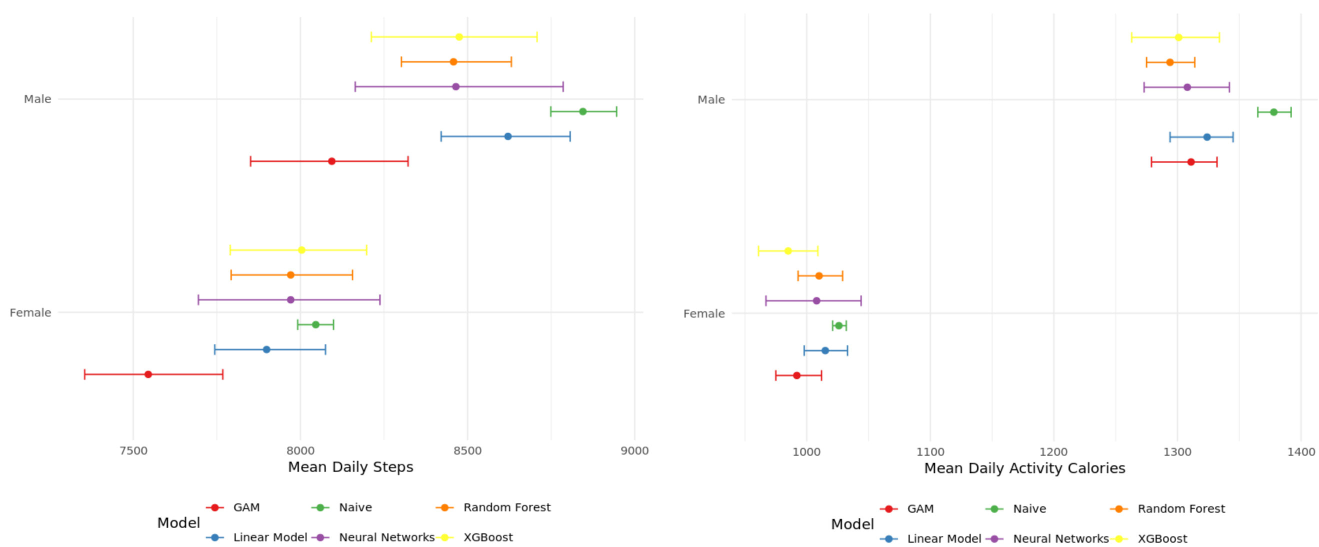


Figure 3.10: Transported gender sub-domain estimates for daily steps (left) and activity calories (right) generated via the Matched Mass Imputation (MMI) framework.

Comparative Model Stability Across Frameworks

The behavior of the predictive architectures shifted noticeably between the two integration frameworks (Figure 3.6). Under Standard Mass Imputation, the linear regression model acted as a directional outlier by overestimating daily steps at 8,502, well above the machine learning consensus. However, in the Matched Mass Imputation framework, the linear model

(8,258 steps) moved directly into the center of the machine learning consensus (8,236–8,263). Conversely, the Generalized Additive Model (GAM) emerged as a potential outlier in both frameworks but showed increased divergence in the Matched framework. In the MMI results, the GAM produced a step estimate of 7,817, which was more than 400 steps lower than the consensus of the other architectures. These interactions suggest that while demographic matching can stabilize simpler models like linear regression, it may introduce new sensitivities in more flexible architectures like the GAM.

Chapter 4

Discussion

Reverse Selection Bias in the 65+ Demographic

The discovery that transported national estimates for the 65+ demographic are higher than the Naive volunteer averages suggests a unique reverse-selection bias in older populations. While younger participants in digital health studies typically represent a healthy user subset and exhibit higher activity levels than the general population, older adults in the All of Us source pool appear to underrepresent the true activity levels of the general U.S. elderly population. This finding challenges the stereotype of the sedentary senior and emphasizes the necessity of using representative frameworks to set national health benchmarks. It suggests that digital health studies may unintentionally reach a more sedentary subset of the elderly while failing to capture the active, community-dwelling seniors who remain mobile and independent.

Metric Resilience: Steps versus Calories

A primary finding of this project is that selection bias impacts daily steps more significantly than activity calories. This divergence suggests a compensatory effect between movement volume and body mass, where the general U.S. population may take fewer steps than study volunteers but likely possesses a higher average body mass. Because energy expenditure is partially a function of body mass, the calorie requirements of the national population remain high despite a lower volume of movement. This indicates that while the U.S. population is more sedentary than the volunteer cohort, the total physiological work performed remains comparable because of the increased energy cost associated with moving a larger body mass.

This resilience is further validated by the framework comparison. Even as the Matched framework identified a higher volume of daily steps, the calorie estimates remained consistently stable or moved slightly lower across both methodologies. This suggests that the imputation frameworks are effectively decoupling movement volume from total energy expenditure. Rather than assuming a fixed linear relationship where calories always follow step trends, the models are capturing a more nuanced biological profile that accounts for the distinct demographic and physical characteristics of the national population.

Demographic Disparities in Physical Activity

The sub-domain analysis demonstrates that selection bias is not uniform across the American population. The framework revealed that the Black demographic is more active in the general population than is suggested by the Naive volunteer sample, whereas the Hispanic demographic exhibited a significant downward adjustment in movement volume. These corrections suggest that current wearable research may be mischaracterizing the activity profiles of minority groups, potentially leading to public health interventions that are misaligned with the actual needs of these communities. Furthermore, the stark contrast between younger cohorts and the 65+ demographic reveals that age is a primary driver of selection bias because these groups were adjusted in entirely opposite directions. Establishing these corrected baselines is a critical step toward ensuring that digital health move from being a majority-representative tool to one that offers equitable insights for all demographic brackets.

Validation Through Framework Comparison

The use of two distinct methodologies, including Standard Mass Imputation and Matched Mass Imputation, provided a rigorous internal validation of these results. Both frameworks independently identified the downward shift in steps and the upward shift in the 65+ subdomain. The Matched framework consistently yielded higher daily step estimates than the Standard version, which indicates that the MMI approach may prevent the over-correction of activity levels. By restricting the source pool to demographically aligned matches, the Matched framework acts as a vital safety check against potential biases in the integration process. The convergence of these two methods across different architectures provides high confidence in the reliability and stability of the national estimates.

Beyond directionality, the MMI framework significantly improved the stability of the estimates. While the machine learning models in the Standard framework were spread across a range of 165 steps, the MMI framework caused the models to converge within a remarkably tight 27-step range. This reduction in variance suggests that demographic matching not only produces more conservative estimates but also creates a more stable predictive environment for ensemble architectures.

Model Robustness and the Case for Consensus

The high level of agreement between the various machine learning architectures provides strong evidence for the robustness of these national estimates. In the Standard framework, the divergence of simple linear regression from the machine learning consensus suggests that the relationship between demographic variables and physical movement is primarily non-linear, requiring the nuance offered by ensemble models like XGBoost and Random Forest. However, the finding that the linear model improved its alignment under the Matched framework suggests that source selection through MMI can mitigate some of the inherent limitations of simpler architectures. At the same time, the divergence of the GAM in the matched subset suggests that highly flexible models may be more sensitive to the specific distributions created by matching. These interactions reinforce the value of identifying a model consensus rather than relying on a single algorithm. By looking for agreement across multiple architectures, this study ensures that national estimates are not skewed by the unique sensitivities of any one model to the underlying data structure.

Chapter 5

Conclusion and Impact

Summary of Findings

This research successfully established sensor-based national estimates for physical activity by integrating wearable data from the All of Us Research Program with the NHANES probability sample to correct for inherent selection bias. By employing a multi-model consensus approach, the study identified a robust national benchmark range of 8,124 to 8,289 daily steps and 1,161 to 1,180 activity calories under the Standard framework. The application of Matched Mass Imputation (MMI) significantly enhanced the stability of these estimates, tightening the machine learning consensus for steps to a highly precise range of 8,236 to 8,263.

A major finding was the identification of a reverse-selection bias in older populations, where corrected national estimates for the 65+ demographic exceeded Naive volunteer averages. This suggests that digital health studies may unintentionally recruit a more sedentary subset of the elderly population while failing to capture more active, community-dwelling seniors. Furthermore, a compensatory effect between movement volume and body mass explains why calorie estimates remained stable despite the downward shift in steps. This indicates that while the general population takes fewer steps than volunteers, their physiological energy expenditure remains comparable due to the increased energy cost associated with a higher average body mass.

Public Health Impact and Health Equity

This project provides a scalable blueprint for public health surveillance by integrating high-resolution wearable data with representative survey data. This approach allows officials to establish objective national benchmarks without the prohibitive cost of device-based national studies. Furthermore, the framework ensures that underrepresented sub-populations are accurately counted in national datasets, which provides a more equitable foundation for exercise and nutrition policy. By ensuring that health guidelines are based on the actual movement patterns of the U.S. population rather than a subset of healthy volunteers, this methodology supports the development of more inclusive public health standards.

Establishing these objective baselines allows organizations such as the CDC to more accurately track long-term health trends and provide a data-driven assessment of the nation's progress toward federal physical activity targets. From a health equity perspective, the sub-domain analysis is perhaps the most critical contribution of this work. Identifying demographic-specific shifts, such as the upward adjustment for the Black demographic and the downward adjustment for males, reveals that selection bias is not uniform across the American population.

Without these corrections, public health guidelines risk being based on a digital divide that mischaracterizes the activity profiles of minority groups and specific age brackets. Accurate sub-domain data allows for more precise resource allocation because identifying specific demographic groups with lower movement volumes helps government agencies better target funding for community fitness programs and localized health interventions. Furthermore, these data provide a vital metric for the USDA and healthcare economists. Accurate calorie averages allow for the calculation of the national energy gap, which is the difference between caloric intake and expenditure, essential for tailoring nutritional guidelines and predicting future healthcare spending related to metabolic health. On a clinical level, these findings allow providers to set evidence-based goals for patients that reflect real-world national and group-specific averages rather than arbitrary marketing numbers. Even at the corporate level, these benchmarks provide a standard against which companies can justify investments in workplace wellness programs by comparing employee activity to the national and demographic norm.

Limitations

Several limitations should be considered when interpreting these results. The validity of the transportability relies on the assumption of Strong Ignorability, which assumes that the models have accounted for all major factors that influence both study participation and physical activity levels. While the harmonized predictors are strongly associated with movement, unobserved factors such as occupational demands or specific device adherence patterns may still influence outcomes and remain unaccounted for. Additionally, the reliance on complete case analysis required the exclusion of participants with missing data. While weight recalibration was used to maintain the demographic balance of the sample, this listwise deletion remains a limitation because it assumes that the data is missing at random. If the individuals excluded for missing data differ systematically in their activity levels from those retained, the final estimates may still contain residual bias. Finally, as these are model-based projections rather than direct measurements within NHANES, the results reflect the transportable relationships derived from the volunteer cohort rather than primary observations.

Future Work

This study establishes a scalable blueprint for data integration, but several avenues for methodological and conceptual refinement remain. Future iterations of this framework could implement formal hyperparameter optimization, such as Bayesian search, to further refine the predictive accuracy of the ensemble architectures. Additionally, while 1:1 nearest-neighbor matching was effective for creating common support, exploring alternative strategies like Propensity Score Matching or Mahalanobis distance with varied ratios could offer further insights into model stability. Beyond algorithmic adjustments, the feature space should be expanded to include Social Determinants of Health (SDoH) and geospatial data to better account for environmental influences on activity. Transitioning from listwise deletion to Multiple Imputation would also allow for the inclusion of participants with incomplete records, strengthening the power of sub-group analyses. Ultimately, as the All of Us Research Program continues to evolve, this methodology can be applied to other longitudinal metrics—such as sleep architecture and heart rate variability—providing a continuous, objective update to the national health profile.

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Appendix: Technical Specifications

A.1 Software Environment

The integration and transportability frameworks were implemented using R (v4.1) within a Jupyter Notebook environment hosted on the *All of Us* Researcher Workbench. The following libraries were utilized for data orchestration, nearest-neighbor matching, and the generation of imputed national estimates:

- **Data Management:** `tidyverse` (v1.3.1) for data cleaning and `nhanesA` (v0.8.2) for retrieving NHANES probability sample data.
- **Nearest-Neighbor Matching:** `FNN` (v1.1.3) was utilized during the Matched Mass Imputation process to restrict the source pool to the most demographically similar volunteers.
- **Predictive Imputation Models:** `xgboost` (v1.5.0), `randomForest` (v4.6-14), `mgcv` (v1.8-38), and `nnet` (v7.3-16).

A.2 Model Hyperparameters

To ensure the stability of the model consensus, the following configurations were utilized across both frameworks. These parameters were selected to balance predictive accuracy with conservative regularization to prevent overfitting:

- **xgboost:** Both models used 200 rounds, a learning rate of 0.05, and a subsample ratio of 0.8. The daily steps model utilized a `max_depth` of 3 with high regularization (`lambda`=5, `alpha`=1, `min_split_loss`=10), while the calorie model utilized a `max_depth` of 4 with more moderate regularization (`lambda`=0.5, `alpha`=0.5, `min_split_loss`=2).
- **randomForest:** 500 trees were grown with an `mtry` of 2 and a `nodesize` of 100 to ensure stable terminal nodes across the volunteer cohort.
- **mgcv (GAM):** Thin plate regression splines were applied to age, height, weight, active minutes, and sedentary standards using a basis dimension of $k = 5$.
- **nnet (Neural Network):** Both architectures utilized a single hidden layer of size 4 with linear output (`linout=TRUE`). The steps model used a decay of 0.2 over 1,000 iterations, while the calorie model used a decay of 0.1 over 2,000 iterations.

A.3 Survey Weight Recalibration

Because the listwise deletion of records with missing predictors can alter the survey's original population totals, a weight recalibration was performed to maintain the integrity of the national estimates. The adjusted weight $w_{i,adj}$ for each participant i in the analytical sample

(S_{ana}) is calculated by applying an adjustment factor κ to their baseline examination weight, $w_{i,base}$:

$$w_{i,adj} = w_{i,base} \times \kappa, \quad \text{where} \quad \kappa = \frac{\sum_{j \in S_{full}} w_{j,base}}{\sum_{k \in S_{ana}} w_{k,base}}$$

In this formulation, $w_{i,base}$ represents the original survey weight assigned to an individual by NHANES prior to any study-specific filtering. The adjustment factor κ is defined as the ratio of the population sum from the full demographic file (S_{full}) to the sum of baseline weights for only those participants remaining in the final analytical sample (S_{ana}). By scaling each individual's weight by κ, the analytical sample is forced to sum back to the original census-based population total. This ensures that the mass imputation results accurately reflect the intended demographic proportions of the U.S. population despite the reduction in sample size caused by missing data.